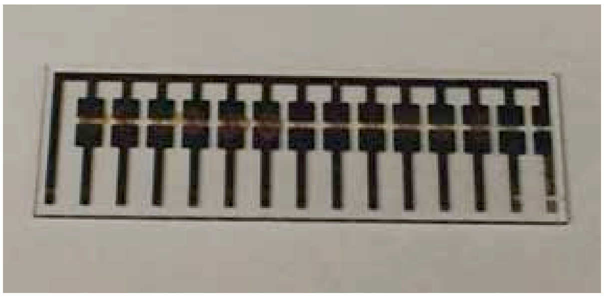
perhaps even replace cell organelles in unhealthy cells, and in the very very long term, repair the telomeres that cap our DNA (whose wear has been directly linked with aging). Nanosensors open up a whole new paradigm. Not only can they revolutionise medicine with an unprecedented ability to interact one-on-one with viruses, but they will also induce huge strides in materials science research.



Source: http://news.mit.edu/2015/sensor-detects-spoiled-meat-0415

This device, based on modified carbon nanotubes, can detect amines produced by decaying meat.

The thought of little robots swimming around in our bloodstreams might induce panic in some, and practical questions in others. How are these devices going to be powered? What happens when they wear out or need maintenance? The answers lie respectively in self-powered and biodegradable sensing technologies.

Self-powered sensors

Research in self-powered sensing is throwing up all kinds of innovative answers. Work is currently being done on sensors that can derive energy from a heat difference (such as that between a patient's body and the surrounding air), from radiation, from chemical energy in compounds like stomach acids, and even from vibrations (such as the beating heart, or thermal currents in the oceans and skies). As always, research is never unidirectional, and the potential for self-powered sensing is being extended to almost any application where it would be hard to keep replacing batteries. Underground vibration monitoring for earthquake and fault detection, especially in high-risk areas around fracking oilfields, is important.

Biodegradable sensors

The concept of biodegradable sensing offers a simple conundrum - the more complex a part, the harder it is to ensure that it disintegrates harmlessly. The idea is to print circuitry on sheets of silicon so thin that they dissolve in water over a period of days to weeks. The circuitry is wrapped in silk to vary this time period - progress has already been made with the fabrication of diodes as well micro-cameras of this type. The possible applications of such sensors are endless. Post-operative care for sensitive organs like the brain and the heart often involve the insertion of sensors into these organs, which not only must be removed at a later date, but can also cause complications, swellings and infections. The advent of innocuous biodegradable sensors stands to completely change this. The potential to non-intrusively collect data on the functioning of natural ecosystems, especially sensitive ones like coral reefs and rivers, also stands to gain.

Vibration and dust/particle sensors

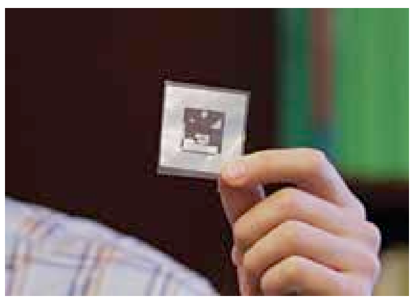
Vibration sensing research has advanced to the extent that it is now commonly used in both industry and consumer applications. The running of huge, hundreds of kiloWatt-hour motors can throw up unexpected problems of alignment that quickly spiral out of control, permanently damaging often expensive equipment. Vibration sensors, especially in their modern avatars that provide compact, comprehensive sensing of amplitude, frequency and wave modes, allow such catastrophes to be avoided in the pumping, automotive, manufacturing and power generation industries. Vibration sensing also has obvious applications in motion sensing and security. Snakes, commonly thought to have no sense of hearing because they lack an external ear, can actually sense vibrations very accurately via an organ in their mouths. This allows them to pinpoint with remarkable accuracy whenever anything is moving around them. Particle sensing equipment has been a focus of sensor research with fairly straightforward applications. Quality control in factories, air conditioning applications, effluent analysis for toxins and particulates, and radiation monitoring are now all processes that can be easily and efficiently automated.

Chemical Reactant Sensors

While it doesn't look like we're going to have anything to compete with dogs for their olfactory abilities in the near future, it certainly behooves humans to try their luck. And so we see chemical reactant sensors, designed as electronic noses that react with specific compounds in the air to detect biohazards, gas leaks, rust and even alcohol on the breath!

Moisture sensors

Moisture sensing technology, though rooted in the mid 1900s, has come a long way - it no longer has to be replaced as often, offers a long life, low maintenance, and excellent accuracy. For certain kinds of post-



Source: http://news.mit.edu/2014/wireless-chemical-sensor-for-smartphone-1208

A wireless chemical sensor.

operative care, it's extremely important to keep a patient well-hydrated in a humid environment (a moist wound heals faster). Moisture sensing helps out here. Also, with integration with other sensors, smart refrigerators can decide you shopping list for you and modify the controls according to what's inside.